

Missing Data

120 minutes · 6 self-check questions

Jesse Anak · Institute of Computational Biology and Bioinformatics

Why data are missing determines what you may do about it. Deletion, single imputation and multiple imputation, and what each assumes.

Learning objectives

- Quantify and display the pattern of missingness in a dataset
 - Distinguish MCAR, MAR and MNAR and explain why the distinction matters
 - State the cost of listwise and pairwise deletion
 - Explain why mean substitution understates uncertainty
 - Describe what multiple imputation does and when it is appropriate
 - Report missing data handling so a reader can judge it
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Look before you decide

- Report the amount missing per variable, not just overall
 - Look at the pattern: is missingness concentrated in a few participants, or spread thinly?
 - A variable missing for 40% of participants is a different problem from 40 variables each missing 1%
 - In R: `colMeans(is.na(dat))` gives the proportion missing per variable
 - Compare respondents and non-respondents on the variables you do have — this is the evidence for whether missingness is random
 - Item non-response (a skipped question) and unit non-response (a participant who never took part) are different problems
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Three mechanisms

- MCAR — Missing Completely At Random: missingness unrelated to anything, observed or not. A dropped test tube.
- MAR — Missing At Random: missingness explained by variables you observed. Older participants skip the income question more often, and you recorded age.
- MNAR — Missing Not At Random: missingness depends on the unobserved value itself. The highest earners decline to state income.
- MCAR is testable in part; MAR is an assumption; MNAR cannot be verified from the data at hand

- Most real missingness is MAR at best, which is precisely the case imputation is built for
- MNAR requires sensitivity analysis under different assumptions rather than a single fix

WHAT EACH MECHANISM PERMITS

Mechanism	Example	Deletion gives	Preferred handling
MCAR	Sample lost in transit	Unbiased, less precise	Complete cases acceptable
MAR	Older people skip income	Biased	Multiple imputation
MNAR	High earners refuse income	Biased	Sensitivity analysis; model the mechanism

Deletion

- Listwise (complete case) deletion drops any participant missing any analysis variable — the default in most software
- With 10 variables each 5% missing, complete cases can lose 40% of the sample
- It is unbiased only under MCAR; under MAR it biases estimates
- Pairwise deletion uses all available data for each computation, so different coefficients rest on different subsamples
- Pairwise can produce a correlation matrix that is mathematically impossible, which breaks regression and factor analysis
- Deletion is defensible when missingness is genuinely tiny — say under 5% and plausibly MCAR — and you say so

IN PRACTICE

Report how many participants each analysis actually used. A table saying $n = 600$ in the methods and a regression silently fitted on 380 is one of the easiest things for a reviewer to catch.

Imputation

- Mean substitution fills gaps with the variable mean; it preserves the mean, shrinks the variance, and biases standard errors downwards
- Because it pretends imputed values are as certain as observed ones, it manufactures significance
- Regression imputation predicts the missing value from other variables — better, but still too certain
- Multiple imputation creates several completed datasets, analyses each, and pools the results, so the uncertainty of imputation is carried into the standard errors
- Rubin's rules do the pooling; 5-20 imputations is typical, more if missingness is heavy
- In R: `mice(dat, m = 20) then pool(with(imp, lm(y ~ x)))`

- Include the outcome and any variable predictive of missingness in the imputation model, or you bias towards the null
- Last observation carried forward is not recommended for longitudinal data — it assumes no change, which is rarely true

Reporting

- State the amount missing per variable
- State the assumed mechanism and the evidence for it
- State the method used and the number of imputations if applicable
- State the analysis sample size for every model reported
- Where missingness is substantial, present a complete-case analysis alongside as a sensitivity check
- A reader should be able to judge whether your conclusion could plausibly be an artefact of the missing data

Worked examples

How 5% missing becomes 40% lost

A regression uses 10 variables, each with about 5% missing values, roughly independently. The dataset has 600 participants. The output says $n = 358$.

1. SPSS and R both delete any case missing any variable in the model — listwise deletion, silently.
2. The chance a participant is complete on all ten is 0.95^{10} , about 0.60.
3. 600×0.60 is 360, which matches the 358 reported.
4. So 40% of the sample was discarded without a warning appearing anywhere.
5. Worse, if missingness relates to age or income, the remaining 358 are not a random subset — the estimates are biased, not just imprecise.
6. Multiple imputation with `mice(dat, m = 20)` uses all 600 and carries the imputation uncertainty into the standard errors.

THE POINT

Always report the analysis n for every model, and compare it with the recruited n . A gap that large should be explained, not left for a reviewer to find.

Why the outcome goes into the imputation model

A student imputes missing income values but deliberately leaves the outcome — depression score — out of the imputation model, reasoning that including it would be circular.

1. The instinct is understandable but the statistics say the opposite.
2. Imputation is not prediction of the truth; it is generating values with the right joint distribution.
3. Omitting the outcome imputes incomes that carry no relationship to depression.
4. Those imputed cases then dilute the real association, biasing the estimate towards the null.
5. Include the outcome, and any variable that predicts missingness, in the imputation model.
6. Include more variables in the imputation model than in the analysis model — that is the recommended practice, not a mistake.

THE POINT

Leaving the outcome out feels cautious and is actively harmful. It is one of the most common imputation errors, and it always pushes results towards finding nothing.

Key terms

MCAR

Missing Completely At Random — missingness unrelated to observed or unobserved values.

MAR

Missing At Random — missingness explained by observed variables.

MNAR

Missing Not At Random — missingness depends on the unobserved value itself.

Listwise deletion

Excluding any case with a missing value on any variable in the analysis.

Multiple imputation

Creating several plausible completed datasets and pooling the results, so imputation uncertainty is reflected in the standard errors.

Rubin's rules

The formulae for combining estimates and standard errors across multiply imputed datasets.

Activity

For your own dataset, tabulate the percentage missing for every variable. Pick the worst one and argue, from what you know about how the data were collected, whether it is plausibly MCAR, MAR or MNAR — and say what that implies for how you should handle it.

Further reading

- **Sterne et al. (2009)** — "Multiple imputation for missing data in epidemiological and clinical research" — BMJ, open access, and the standard practical reference.
- **van Buuren** — **Flexible Imputation of Missing Data** — Free online; author of the mice package.
- **Little & Rubin** — **Statistical Analysis with Missing Data** — The theoretical foundation, if you need it.

Self-check questions

1. Older participants are more likely to leave the income question blank, and you recorded everyone's age. This missingness is:
 - a. MCAR
 - b. MAR
 - c. MNAR
 - d. Not missing data at all
2. The people who decline to state income are those with the highest incomes. This is:
 - a. MCAR
 - b. MAR
 - c. MNAR
 - d. Listwise deletion
3. Replacing missing values with the variable mean primarily causes:
 - a. The mean to shift
 - b. The variance to shrink and standard errors to be too small, overstating significance
 - c. The sample size to fall
 - d. No problem, provided under 10% is missing
4. Ten variables each have 5% missing, roughly independently. Complete-case analysis will retain approximately:

- a. 95% of the sample
 - b. 60% of the sample
 - c. 50% of the sample
 - d. 5% of the sample
5. The essential advantage of multiple imputation over single imputation is that it:
- a. Produces a larger sample
 - b. Carries the uncertainty of the imputation into the standard errors
 - c. Requires no assumptions
 - d. Works even when data are MNAR
6. The outcome variable should be included in the imputation model because omitting it:
- a. Makes the imputation run more slowly
 - b. Biases associations towards the null by imputing predictors as unrelated to the outcome
 - c. Is forbidden by Rubin's rules
 - d. Has no effect
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Answers

- 1. **b.** Missingness depends on age, which you observed. That is MAR — and it is exactly the condition under which multiple imputation performs well, because the imputation model can use age.
 - 2. **c.** Missingness depends on the unobserved value itself. No amount of imputation from observed variables fixes this; it calls for sensitivity analysis under different assumptions about the missing values.
 - 3. **b.** Mean substitution leaves the mean unchanged but piles values at the centre, shrinking variance. Treating imputed values as if they were observed manufactures precision the data do not support.
 - 4. **b.** 0.95 to the power of 10 is about 0.60. Small amounts of missingness across many variables compound, which is why complete-case analysis so often discards far more data than researchers expect.
 - 5. **b.** Multiple completed datasets, analysed separately and pooled, propagate the fact that the imputed values were estimates. Single imputation treats a guess as data.
 - 6. **b.** Leaving the outcome out imputes predictor values that carry no relationship to it, diluting the very association you are estimating. It feels like circularity but is the correct procedure.
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